IILM UNIVERSITY

School of Computer Science and Engineering

Internship Report Presentation

Cybersecurity & Ethical Hacking

STUDENT NAME	Aman Singh Tanwar
ENROLLMENT NUMBER	CS-23411126
PROGRAM & SEMESTER	B.Tech CSE, VII Semester
ORGANIZATION	ApexPlanet Software Pvt. Ltd.

ABOUT THE ORGANIZATION

ApexPlanet Software Pvt. Ltd.

- Delivers secure, high-performance web and mobile applications by blending modern engineering with cybersecurity best practices.
- Builds cyber-resilient solutions using secure coding, encryption, and penetration testing to protect businesses from evolving threats.
- Trains the next generation of developers in security-first engineering through hands-on, industry-relevant internship programs.
- Website: www.apexplanet.in

Internship Overview

DOMAIN

Cybersecurity & Ethical Hacking

DURATION

8 Weeks & 3 Days (60 Days)

TIMELINE

21 Apr 2026 – 19 Jun 2026

MODE

Virtual / Online

Objective

Build strong fundamentals in cybersecurity, networking and cryptography, and set up a professional hacking lab to progressively learn reconnaissance, scanning, web application security, exploitation, system hardening, and incident response through five structured tasks.

Program Structure — 5 Tasks

T1
Foundations of Cybersecurity

T2
Network Security & Scanning

T3
Web Application Security

T4
Exploitation & System Security

T5
Capstone: Mini SIEM (ELK) & Incident Response

T1

Foundation & Environment Setup

TASK 1 | Days 1–12

OBJECTIVE

Build strong fundamentals in cybersecurity, networking, and cryptography, and set up a professional hacking lab.

KEY STEPS

- Cybersecurity basics: CIA Triad, threat types, and attack vectors
- Lab setup: VirtualBox/VMware, Kali Linux, Metasploitable2, DVWA
- Linux fundamentals: navigation, permissions, package management
- Networking basics: OSI model, TCP/IP, DNS/HTTP, subnetting & NAT
- Cryptography basics: encryption, hashing, SSL/TLS, OpenSSL
- Tool familiarization: Wireshark, Nmap, Burp Suite, Netcat

DELIVERABLES

- ✓ Lab Setup Report with screenshots
- ✓ GitHub repo with notes & Linux cheat-sheet
- ✓ 5-min video walkthrough of lab setup

T2

Network Security & Scanning

TASK 2 | Days 13–24

OBJECTIVE

Learn reconnaissance, scanning, and network traffic analysis.

KEY STEPS

- Passive & active recon: Whois, Nslookup, Google Dorking, Shodan
- Port & service scanning with Nmap (TCP/UDP, version & OS detection)
- Vulnerability scanning with OpenVAS on Metasploitable2
- Packet analysis with Wireshark (HTTP, FTP, DNS traffic)
- SYN flood simulation and analysis using hping3
- Firewall basics: iptables rules and blocking port scans

DELIVERABLES

- ✓ Nmap Scan Report + OpenVAS Vulnerability Report
- ✓ GitHub repo with detailed scan analysis
- ✓ 5-min demo video showing scan & findings

T3

Web Application Security

TASK 3 | Days 25–36

OBJECTIVE

Identify and exploit OWASP Top 10 vulnerabilities in a controlled lab environment.

KEY STEPS

- SQL Injection on DVWA + prevention via prepared statements
- Cross-Site Scripting (XSS): stored & reflected, with CSP mitigation
- Cross-Site Request Forgery (CSRF) and token-based protection
- File inclusion attacks: local (LFI) and remote (RFI)
- Burp Suite: intercepting requests and fuzzing with Intruder
- Web security headers analysis via securityheaders.com

DELIVERABLES

- ✓ Security Testing Report (SQLi, XSS, CSRF) with screenshots
- ✓ GitHub repo with attack scenarios + mitigation notes
- ✓ 8-min video demo of exploitation & fix

T4

Exploitation & System Security

TASK 4 | Days 37–48

OBJECTIVE

Learn the penetration testing workflow and exploit vulnerabilities responsibly.

KEY STEPS

- Pentest methodology: Recon → Scan → Exploit → Post-Exploit → Report
- Exploitation with Metasploit on Metasploitable2, reverse shell access
- Password attacks: SSH brute-force (Hydra), hash cracking (John the Ripper)
- Social engineering simulation: phishing awareness page (simulation only)
- Malware basics: static vs dynamic analysis in a sandbox
- System hardening: patches, firewall rules, disabling unused services

DELIVERABLES

- ✓ Penetration Testing Report with screenshots
- ✓ GitHub repo documenting steps + mitigations
- ✓ 10-min demo video of an exploit with explanation

T5

Capstone Project & Incident Response

TASK 5 | Days 49–60

OBJECTIVE

Capstone chosen: Build a Mini SIEM (Security Information & Event Management) using the ELK Stack to centrally monitor, detect, and respond to simulated attacks.

KEY STEPS

- Set up Elasticsearch, Logstash & Kibana (ELK Stack) on a dedicated VM
- Deploy Filebeat on Kali, Metasploitable2 & DVWA to ship logs centrally
- Build Logstash pipelines to parse SSH, web-server & scan activity logs
- Create Kibana dashboards to visualise failed logins, scans & web attacks
- Configure alerts for brute-force attempts and port-scan patterns
- Re-run Hydra/Nmap attacks, detect them live, then contain & document via a post-incident report

DELIVERABLES

- ✓ Mini SIEM Capstone Report PDF
- ✓ GitHub repo with ELK configs, pipelines & dashboards
- ✓ 12-min video demo of SIEM detecting & responding to the attack

Tools & Technologies Used

1 Lab & OS

- VirtualBox / VMware
- Kali Linux
- Metasploitable2, DVWA

2 Recon & Scanning

- Nmap
- Whois, Nslookup, Shodan
- OpenVAS / Nessus

3 Traffic & Web

- Wireshark
- Burp Suite
- securityheaders.com

4 Exploitation

- Metasploit Framework
- Hydra
- John the Ripper

5 Crypto & Hardening

- OpenSSL
- iptables
- hping3

6 SIEM & Docs (Capstone)

- ELK Stack: Elasticsearch, Logstash, Kibana
- Filebeat (log shipping)
- GitHub

Key Learnings & Outcomes

- Strong practical grasp of cybersecurity fundamentals, networking, and cryptography.
- Hands-on proficiency in building and working within an isolated pentesting lab.
- Practical experience in reconnaissance, scanning, and vulnerability assessment.
- Ability to identify, exploit, and mitigate OWASP Top 10 web vulnerabilities.
- Familiarity with end-to-end penetration testing methodology using Metasploit.
- Understanding of system hardening and the basics of incident response.
- Improved technical documentation, reporting, and presentation skills.

Internship Completion Certificate

CERTIFICATE OF COMPLETION

INTERNSHIP PROGRAM

This is to certify that

AMAN SINGH TANWAR

has successfully completed the **Internship Program** at **ApexPlanet Software Pvt Ltd** in the domain of **CYBERSECURITY & ETHICAL HACKING** from **21 April, 2026** to **19 June, 2026** under the guidance of ApexPlanet Software Pvt. Ltd.

Duration: 8 Weeks & 3 Days | Mode: Virtual/Online

During the internship period, the intern actively participated in assigned tasks, project development activities, and practical learning exercises.

We appreciate the dedication and efforts shown throughout the internship and wish them success in future endeavors.

Authorized Signatory
Founder & CEO, ApexPlanet Software Pvt. Ltd.

Certificate ID: APSPL2632834
Scan the QR code to verify this certificate

AICTE

www.apexplanet.in | Empowering Future Tech Leaders

Thank You

Aman Singh Tanwar | CS-23411126 | B.Tech CSE, VII Semester

Cybersecurity & Ethical Hacking Internship • ApexPlanet Software Pvt. Ltd.